Plain Language Explanation of Angular Manifold Routing

Modern AI systems are powerful but extremely expensive to run. This is because most models perform large amounts of computation even when much of it is unnecessary.

Angular Manifold Routing is a new method that allows AI systems to organize inputs geometrically so that many of them can share the same compute resources.

The key discovery is that when embeddings are normalized and routed only by direction, inputs naturally cluster into a smaller number of routing groups.

$E(K) \approx c \cdot K^{0.572}$

This means compute grows slower than the available routing capacity, producing natural sparsity.

Experiments show this can reduce effective routing cost by 3–5× and significantly improve hardware efficiency.

The approach relies purely on geometry: embeddings lie on a hypersphere and are routed according to angular sectors.

Because related inputs tend to point in similar directions, they naturally cluster together.

The result is a system that performs computation only where it is needed, potentially allowing large AI models to run on less hardware.